CampusHelp AI: A Reproducible Synthetic-Data Pilot for Routing Student Support Questions

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Abstract

Students may be unsure which office to contact for help. This pilot studies whether short student questions can be assigned to five support categories using lightweight text classification. A templated training dataset contains 250 artificial messages. On a stratified 50-example holdout, a keyword baseline attained 94% accuracy while TF–IDF with Naive Bayes and Logistic Regression each attained 100%. On an independent, manually authored 60-question challenge set, accuracies fell to 60%, 60%, and 58.3%, respectively. A local Gradio demonstrator translates predicted categories into generic office suggestions and flags low-confidence results. The divergence reveals weak transfer across synthetic writing styles; neither dataset measures real student performance.

CCS Concepts

• **Computing methodologies** → *Machine learning approaches.*

Keywords

Student support, Intent classification, Text classification, Synthetic data

1 Introduction

University support requests involve academic advising, technology, financial aid, housing and other campus services. Misrouted requests may delay assistance. Our question is: *How do three lightweight classifiers perform on templated synthetic questions versus independently worded, hypothetical student requests?* We implement a deterministic data generator, fixed-split evaluation, three routing methods and a local prototype. The contribution is the application-specific comparison and integrated demonstrator, *not* the invention of Naive Bayes, Logistic Regression, TF–IDF or Gradio. This is supervised multiclass intent classification, not a deployed campus service.

2 Related Work

Larson et al. [1] examined intent classification alongside out-of-scope detection in the CLINC150 dataset and found out-of-scope recognition challenging. Their broader, crowdsourced benchmark motivates caution when evaluating our closed-set, synthetic labels. Joulin et al. [2] showed that efficient shallow models can remain competitive for text classification, motivating lightweight methods rather than an expensive transformer. Zhang et al. [3] explored character-level convolutional alternatives using large-scale datasets; our deliberately smaller experiment does not claim to establish a preferred model across domains. These studies did not evaluate our newly generated campus-routing messages.

Table 1: Accuracy and macro-F1 for the templated holdout ($n = 50$) and independently written challenge set ($n = 60$). The same models were used for both evaluations.

	Holdout		Challenge	
Method	Acc.	F1	Acc.	F1
Keyword baseline	.940	.941	.600	.608
TF–IDF + Naive Bayes	1.000	1.000	.600	.584
TF–IDF + Logistic Reg.	1.000	1.000	.583	.562

3 Methodology

We generated 250 templated questions, 50 each for Academic, Technology, Financial Aid, Housing/Facilities, and Other. Each category combined 10 subject strings with five sentence templates. No student records were collected. Generator seed 42 shuffled examples; a stratified 80/20 split (`random_state=42`) produced 200 training and 50 test records. The baseline counted matching category keywords and defaulted to Other when none matched. For the two learned models, word unigram/bigram TF–IDF features were fitted *only on training data* within scikit-learn pipelines; classifiers were Multinomial Naive Bayes (defaults) and Logistic Regression (`max_iter=2000`, seed 42). No pretrained model or fine-tuning was used. All methods used identical held-out records. We then authored a separate 60-question challenge set (12 per category) featuring novel wording, slang, and some cross-office ambiguity. These prompts were not used to train or tune any classifier, and primary-office labels were assigned during dataset construction by a single annotator, without independent adjudication. Accuracy measures correct classifications; macro-F1 averages per-class F1 equally. Training and testing ran locally with Python and scikit-learn 1.8.0 on CPU. The Gradio interface provides office descriptions and an adjustable low-probability review flag, but no directory integrations. No human participants were recruited or tested, and no IRB determination was sought for the synthetic-only implementation. Sentence-frame similarity across the original split and subjective challenge labels are major limitations.

4 Results

Table 1 records executed predictions on synthetic-only data. The baseline made three errors on the original holdout; both learned models made none. Under the distinct challenge wording, the three approaches correctly labeled 36, 36, and 35 of 60 questions. One Logistic Regression failure was “Where do I appeal a grade that seems wrong?” (expected Academic; predicted Financial Aid). Another was “I missed the add/drop cutoff; who handles an exception?” (expected Academic; predicted Other). Predictions and labels are included with reproducible materials.

5 Discussion and Future Work

The 40–42 percentage-point decline in accuracy of the learned classifiers suggests substantial reliance on repeated phrasing and lexical associations rather than robust intent understanding. Thus, perfect original held-out scores cannot justify campus deployment. The challenge set was deliberately constructed to be difficult, not sampled randomly from students; its performance cannot be interpreted as a population estimate. Some examples span multiple offices, and the single-author labels are contestable. The small dataset, uncalibrated probabilities, lack of out-of-scope evaluation, and generic rather than verified campus offices further limit reliability. The demo's low-confidence flag is implemented but not validated as a safe triage rule. No real student records were collected. Next steps are a consented and appropriately reviewed human-data study, independent label adjudication, group-separated evaluation, calibrated abstention and out-of-scope handling, and verification of routing destinations. None of these extensions were tested in this study.

References

[1] Stefan Larson et al. 2019. An Evaluation Dataset for Intent Classification and Out-of-Scope Prediction. arXiv:1909.02027. https://arxiv.org/abs/1909.02027.

[2] Armand Joulin, Edouard Grave, Piotr Bojanowski, and Tomas Mikolov. 2017. Bag of Tricks for Efficient Text Classification. In *Proceedings of EACL*, 427–431. https://aclanthology.org/E17-2068/.

[3] Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level Convolutional Networks for Text Classification. In *Advances in Neural Information Processing Systems 28*. https://arxiv.org/abs/1509.01626.

Supplemental Page: COMP 365 Final Submission Links

Recorded demo and paper review: *TO BE ADDED BY STUDENT — record screen and paste publicly accessible URL here.*

Reproducible source code and data: *TO BE ADDED BY STUDENT — publish project files to an accessible GitHub repository and paste URL here.*

Live app (optional): *TO BE ADDED BY STUDENT — deploy Gradio to a publicly accessible host and paste URL here.*

Overleaf project: *TO BE SHARED DIRECTLY WITH DR. POPE AND THE TAS.*

Reproduction instructions: Install the versions in `requirements.txt`, run `python build_project.py`, `python evaluate.py`, then `python app.py`. The README explains the dataset provenance and limitations. Verify all published links in an incognito browser before submission.

Research integrity note: This artifact includes newly generated synthetic data; it does not recreate or claim measurements from the earlier checkpoint notebook. The video, public links, instructor feedback, and Overleaf sharing require student action and are not yet completed.